



Example: ASB Electrical System Overview

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Abstract

The Electrical System Document is used to describe the interaction between the Autonomous System Brake (ASB) supervisor and Shutdown Circuit (SDC) on hardware level.



1 Document Requirements

1. The Document shall contain a descriptive section, answering the following questions in a Q&A style, as shown in the example below.
 - How is the SDC latch realized to fulfill T14.5.1 for AS and RES?
 - How does the supervisor open the SDC?
 - How is the SDC opened in case of a supervisor fault?
(e.g. CPU stuck and all IO's remain in their original state)
2. The descriptive section must be text only and properly be referenced to the schematics.
3. The document shall contain a schematic section.
4. Each control unit needed to answer the questions of the descriptive section must be included.
5. For each control unit an overview must be placed before the schematics, which contains all mayor function blocks needed to understand its function, as well as all of its ports in the high level block diagram to enable cross checking of the external connections.
6. Only the schematics needed to understand the answers to the questions above must be shown.
7. For all function blocks that are involved to handle the SDC functions of the descriptive section, a detailed schematic must be presented (all logic between the CPU and the SDC).
8. The schematic must include all ports shown on the overview.
9. The schematics must be clearly structured: signal flow when ever possible left -> right.
10. On digital logic gates, symbols must be used which represents their logical function, e.g. according to IEC 60617-12 or ANSI/IEEE Std 91/91a-1991.
11. Values or part numbers of the components must be shown.
12. The schematics must be embedded with searchable text. You cn use a tool like PDF24 with the "just assemble" option to join multiple PDF pages while maintaining them as vector graphics with text.

2 Review Goals

1. Make sure that the SDC logic is properly implemented.
2. Make sure that the failure detection mechanism can work properly.



3 Review Checklist

1. Are the Document requirements fulfilled and does it fulfill the general style requirements?
2. Is the function correct and understandable from the description and the schematics?
3. Does the latch applies to the CPU introduced opening, the case of failure opening and the RES?
4. Are there no paths, where the CPU can open and close the SDC without latching (at least once the TS got activated)?
5. Are there other reset options then power cycle or a manual reset button?

4 General Document Requirements

1. The document must be printable in DIN A4.
2. Team name, University, Car# must be written on every page.
3. Document name must be written on every page.
4. All pages must be numbered, including total number of pages e.g. Page 1/5.
5. Pictures must be well sized and in good resolution (vector or 300dpi).
6. Relevant text must be embedded as text and not as picture (search-able).

5 Example

Warning: The following example shows only how the document could look like and what is meant by the different checklist points. The technical content is not necessarily compliant to the rules nor necessarily related to the other documents nor function at all.

Electrical System Description

All schematics needed for this description can be found below. Only the orange colored parts in the overview are shown here. The remaining parts are not relevant for this description and thus not shown here.

1 SDC Latch implementation

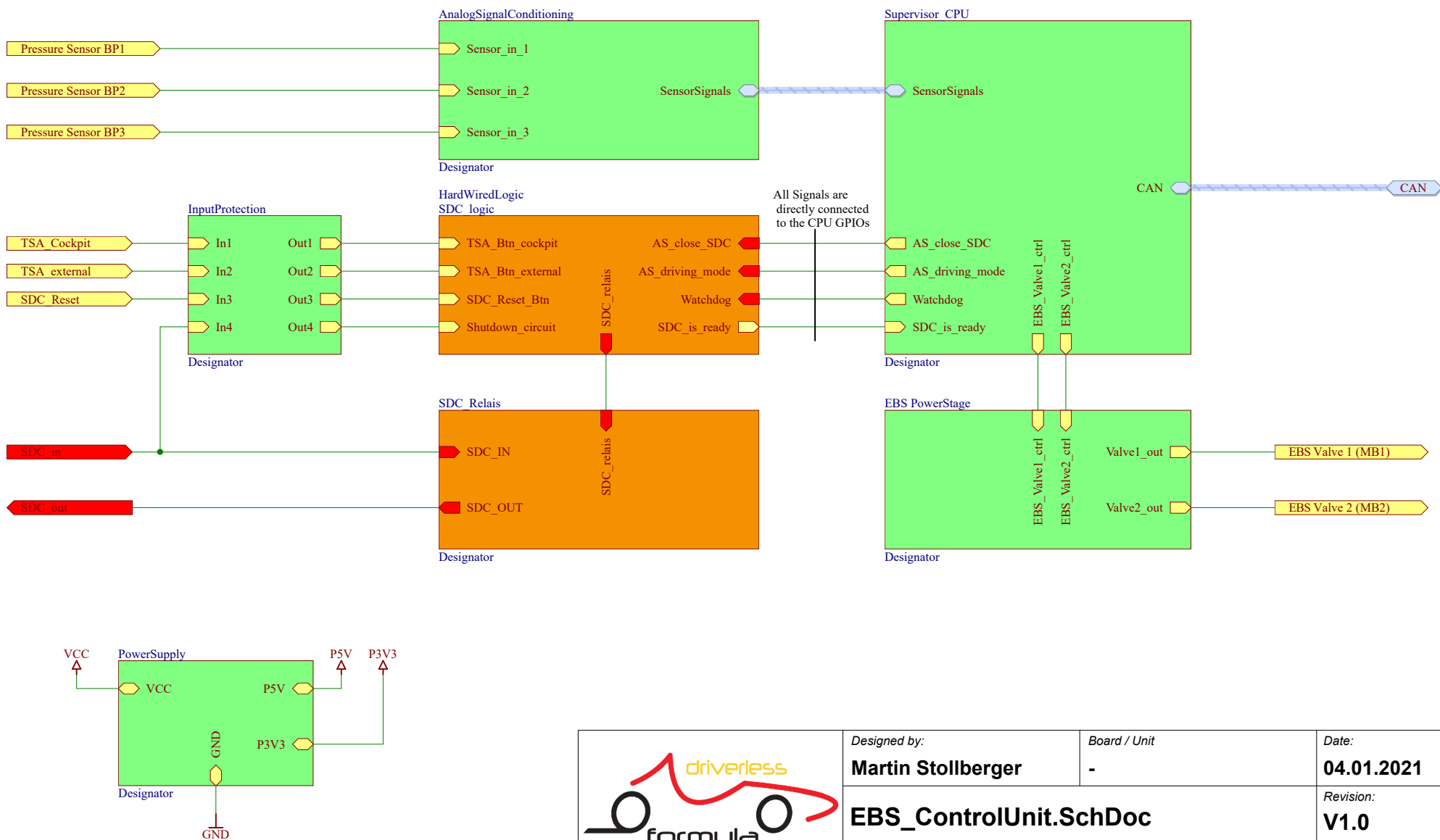
The SDC-Latch function is implemented by using a Positive-Edge-Triggered D-Type Flip-Flop with Clear and Preset (K2). This Flip-Flop can only be set by the reset output of the CPU Supervising chip UCC2946 (U1). This reset output is only set during the power on phase (power on delay = 31 ms after supply rises above 3.0 V) or if the RTH pin is pulled below 1.2 V by V1, which gets activated by the external Reset button. Once the Flip-Flop is set, it opens the AND-Gate K5 and permits the supervisor to close the SDC relays which is connected to K5 via another MOSFET. The closing of the SDC is now done by setting the Flip-Flop K1. Once this Flip-Flop was set it cannot be reset by the CPU and thus cannot reactivate the SDC once more. Setting K1 also enables the path for clearing K2, which is needed to open the SDC again. The SDC opening will be either happen if the shutdown circuit before this circuit is opened, or if U1 issues a watchdog failure.

2 SDC opening by the supervisor

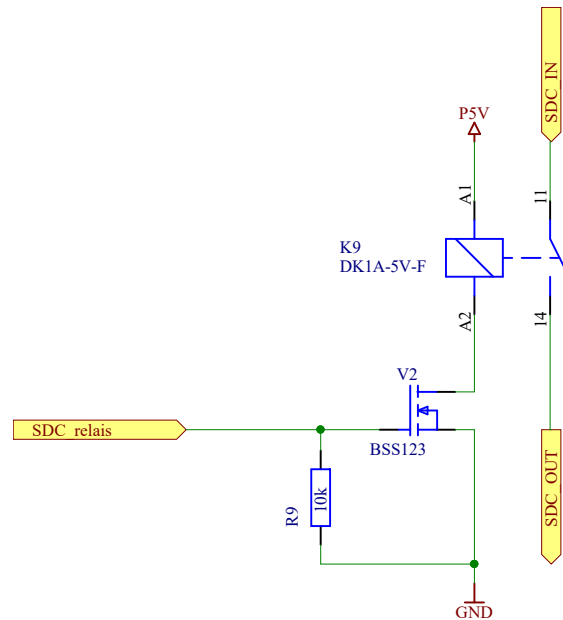
If the supervisor detects a failure in the EBS, it will immediately exit the monitoring task and thus, stops toggling the watchdog pin (see Supervisor Flow Chart). This will lead to an opened SDC as soon as the watchdog timed out after 250 ms. Additionally the EBS actuator power is cut through the EBS power stage to ensure a EBS activation even if the SDC fails.

3 SDC opening on supervisor failure

If a severe failure happens to the CPU itself, if it gets stuck or if the IO pins got stuck, this will automatically lead to a watchdog failure as U1 expects a cyclically toggling signal after 250 ms.



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Changelog

Section	Version	Change
4	V1.0	Initial Version